TALENT NURTURES
NURTURING THE GEN NEXT!!

NCC Global Assignment

NCC education

Qualification: L5DC-BM

Unit Name: Artificial Intelligence

Name of Student: Mhd Laith Alkhateeb

Student NCC ID: 213142

Address: SFO0003, Compass Building, Al Hamra Industrial Zone,

Ras Al Khaima, UAE

Contact No: +971-4-2628591 www.tnedu.org

Assignment Marking Sheet

NCC education® Awarding Great British Qualifications

Centre Name	Talent Nurtures Educational Institute
Candidate Name	Mhd Laith Alkhateeb
NCC Education Candidate ID No.	213142
Examination Date	15/8/2025
Examination Unit Title	DEEP DRIVE

Questions answered by the candidate:

Please insert the question numbers you have answered in the table below.

Question number attempted (To be completed by Candidate)	*Marks Awarded (To be completed by 1st marker)*	*Marks Awarded (To be completed by 2nd marker)*
TOTAL MARKS		

Mhd Laith Alkhateeb ID No: 213142 Word Count: 3621

First Marker Name & Signature: Muhammad Rafiq

Second Marker Name & Signature: **Naseem Gul**

Statement. and .Confirmation .of .Own .Work

NCC education Awarding Great British Qualifications

A signed copy of this form must be submitted with every assignment. If the statement is missing you work may not be marked. For Level 7 assignments, please use the separate Level 7 Candidate statement of own work from instead of this one.

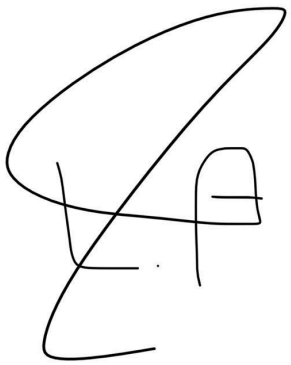
I confirm the following details:

Candidate Name:	Mhd Laith Alkhateeb
Candidate ID Number:	213142
Qualification:	L5DC-BM
Unit:	Artificial Intelligence
Centre:	Talent Nurtures Educational Institute
Word Count:	3621

I have read and understood both NCC Education's Academic Misconduct Policy and the Referencing and Bibliographies document. To the best of my knowledge my work has been accurately referenced and all sources cited correctly.

I confirm that I have not exceeded the stipulated word limit by more than 10%.

I confirm that this is my own work and that I have not colluded or plagiarised any part of it.

Candidate Signature:	
Date:	15/8/2025

Mhd Laith Alkhateeb ID No: 213142 Word Count: 3621

DEEP DRIVE

Mhd Laith Alkhateeb ID No: 213142 Word Count: 3621

Scenario

DeepDrive is a startup business founded in 2019 and it is looking to develop an artificial intelligence system that will enable any truck to be converted into a self-driving autonomous truck. The company has the potential to revolutionise the automotive sector and has received substantial funding from investors all over the world. The current focus of the company is to produce a robust prototype that will be presented at the Consumer Electronics Show (CES) in a year. This is an important milestone since an impressive demo can receive wide media coverage and attract new investors. In the last months, most of the company's effort has been put into developing a mechanical add-on for the control of the steering wheel, gears, and pedals, a set of sensors and cameras, and an electronic board for synchronising those components. Unfortunately, the management has underestimated the effort required to develop an artificial intelligence system for the control software and consequently, the development is lagging behind. You have been hired as the technical lead for the development of the artificial intelligence system that will be used to drive the truck. As a first step, the management has asked you to produce a report in which you provide a detailed plan for the development of the system. Below you will find the instructions for producing this report.

Task 1 – System Overview

This task illustrates the overview of the AI System by describing and comparing the learning paradigms and methods.

a)

> Supervised learning is a category of machine learning that uses labeled datasets to train algorithms to predict outcomes and recognize patterns. Unlike unsupervised learning, supervised learning algorithms are given labeled training to learn the relationship between the input and the outputs.
>
> Supervised machine learning algorithms make it easier for organizations to create complex models that can make accurate predictions. As a result, they are widely used across various industries and fields, including healthcare, marketing, financial services, and more. (cloud.google.com , What is Supervised Learning ?)

Supervised learning is like teaching a student by equipping them with a lot of examples and the answers of these examples. However , humans are incapable of processing huge amounts of examples. Supervised learning can train and equip the AI systems with tons of examples (information) and the corresponding answers to make the AI system predict the output based on the labeled data (answers).

> Reinforcement learning (RL) is a machine learning (ML) technique that trains software to make decisions to achieve the most optimal results. It mimics the trial-and-error learning process that humans use to achieve their goals.
> (aws.amazon.com , What is reinforcement learning ?)

Reinforcement learning is the way of teaching the systems by the mistakes they make. For example, a video game player who plays a new game for the first time would fail in winning because of the lack of knowledge. But by time, the player will learn the rules to avoid making mistakes they did before. Leading to winning games.

Mhd Laith Alkhateeb ID No: 213142 Word Count: 3621

Reinforcement and Supervised Learning Comparison:

Paradigms	Training Time	Computational Costs	Performances	Data Usage
Supervised learning	Time consuming In terms of training Since it depends on large amounts of labeled data. (geeksforgeeks.org, Supervised Machine Learning, 11 Jul, 2025) (disadvantage)	Expensive due to requiring a large amount of labeled data (Disadvantage) (lyzr.ai , Supervised Learning, Dec 3, 2024)	1-Provides accurate predictions (advantage) 2- helps organization to make efficient decisions based on predictions (advantage) (lyzr.ai , Supervised Learning, Dec 3, 2024)	Needs large amounts of labeled data (Disadvantage) (lyzr.ai , Supervised Learning, Dec 3, 2024)
Reinforcement learning	Since it does not require labeled data unlike SL, it learns faster due to its ability to learn from the action it takes. (Advantage) (birchwoodu.org , What are the Reinforcement Learning Advantages and Disadvantages ?, Oct 18, 2024)	Needs substantial amount of data and computational power (Disadvantage) (birchwoodu.org , What are the Reinforcement Learning Advantages and Disadvantages ?, Oct 18, 2024)	1- Ability to solve complex issues. (Advantage) (birchwoodu.org , What are the Reinforcement Learning Advantages and Disadvantages ?, Oct 18, 2024) 2- Ability to learn from real time actions (Advantage) (geeksforgeeks.org , RL, 24 Feb, 2025)	Does not require labeled data as it learns from the actions it takes (Advantage) (birchwoodu.org , What are the Reinforcement Learning Advantages and Disadvantages ?, Oct 18, 2024)

The comparison highlights the different approaches that Supervised Learning and Reinforcement could offer for DeepDrive Plan. Supervised Learning Requires large amounts of labeled data. Which can be expensive and time consuming. However, if collecting the data can be manageable. The training can be done to the system quickly. Additionally, Supervised learning is considered a solid technique in predicting outputs as it provides accurate predictions based on labeled data. Which makes it outstands any other techniques and well suited for situations where the truck must be in a certain lane or sending signs when traffic exists.

On the other hand, Reinforcement learning does not require labeled data as it learns from the actions and mistakes it makes and it is capable of dealing with complex issues efficiently. Which makes it considerable in the long term. However, since it needs a significant amount of data and high computational power. It does not make it ideal for DeepDrive as it may consume time while being trained. Especially that it requires simulations before being qualified for real world performance. Making it unsafe to be produced for the demo by the CES deadline.

b)

“A neural network is a machine learning program, or model, that makes decisions in a manner similar to the human brain, by using processes that mimic the way biological neurons work together to identify phenomena, weigh options and arrive at conclusions.” (ibm.com , what is a neural network?, oct 6, 2021)

“An expert system is a computer program that uses artificial intelligence (AI) technologies to simulate the judgment and behavior of a human or an organization that has expertise and experience in a particular field.” (techtarget.com , expert system, aug 26, 2024)

“A decision tree is a supervised learning algorithm that can be used for both classification and regression tasks. It works by recursively partitioning the data into subsets based on feature values, making decisions at each node to maximize a specific criterion” (e.g., information gain or Gini index). (medium.com , all about decision trees, Sep 12,2024)

Methods	Performance	Possible pitfalls	Type of data they can handle
Neural networks	Very good at handling complex data . (linkedin.com ,advantages and disadvantages of neural network)	poor on new data. (medium.com ,7 common pitfalls in training neural networks and how to avoid them, Sep 20, 2024)	images, videos, text, audio. (aws.amazon.com, what is a neural network ?)
Expert systems	Delivers consistent results, fast. (savemyexams.com, expert systems, 24 may 2024)	Lack of common sense in decision making. (en.wikibooks.org , the computer revolution/ artificial intelligence/ expert systems)	Incompleted data, uncertainty in data, inconsistency in data. (linkedin.com , what are the most effective expert systems for regression analysis?)
Decision trees	Flexible in terms of using both classification and regression tasks (ibm.com , what is decision tree ?)	Poor performance on new data. (medium.com ,limitations of decision trees in supervised learning: an analytical deep dive, nov 25, 2024)	Categorical and numerical data, missing values in datasets. (gyansetu.in, decision trees advantages and disadvantages, may 21, 2025)

Based on the comparison, each method provides advantages and disadvantages for the project of deep drive. Neural networks appear to be the favorable option among the others as it is a powerful method in terms of handling complex data such as in images, texts, and audios. Which can make it a strong choice for the project. However, sometimes it can be inconsistent, time consuming, and expensive due to its reliability on labeled data. which can make it harder to obtain within one year. If used in heavier tasks, making it ready for the demo can be manageable .Additionally, expert systems deliver consistent and fast results when applied to obvious situations such as adhering to speed limits. However, due to the lack of common sense in decision making, it can make it complex to be qualified for real world scenarios.

Decision trees are considered flexible for classification and regression tasks. They can handle Categorical and numerical data such as weather conditions. However, their performance decreases on new data . making them less reliable in real world conditions.

Task 2 – System design

This task showcases the system design

a)

Controller module takes sensor data and a short-term path to create smooth driving commands. Neural networks are the best choice for this. They can handle messy, high volume inputs, like images or distance readings, and turn them into smooth outputs, such as gradual steering adjustments. In autonomous vehicles, neural networks can learn from many driving examples to imitate human-like control and manage unexpected situations, like sudden curves or changing traffic. This approach fits the one year CES demo. They can be trained using existing datasets quickly, with powerful yet affordable computers, without coding every scenario by hand. For instance, studies show that neural networks effectively model steering in passenger vehicles by learning from sensor data. (asco.lcsr.jhu.edu , neural network modeling for steering control of an autonomous vehicle)

On the other hand, decision trees may create choppy, step-like controls because they divide decisions into strict categories. This can lead to jerky rides, which aren't ideal for smooth highway driving. Expert systems might work for simple rules, but they struggle with the endless edge cases in real traffic, making them too fragile and time-consuming to develop within a year.

Next is the Navigation module. It uses a destination to create a route plan on a map. Expert systems excel here because routing relies on simple rules, like avoiding one-way streets, prioritizing highways, and considering speed limits. These can be programmed as if-then statements using map data, making the system reliable and easy to update, like adding detours for construction. This fits perfectly with the demo timeline.

Expert systems do not need large training data; they require input from map specialists, and they are interpretable, allowing engineers to review routes quickly.

Research on transportation shows how expert systems effectively handle vehicle routing and scheduling. (onlinepubs.trb.org , application of expert systems to transportation: a strategy for safety and productivity gains)

Neural networks would be excessive and risky; they might create paths if overtrained on specific patterns. This would require large labeled datasets that we might not collect in time. Decision trees could manage simple choices, but they do not scale well to complex map graphs with thousands of intersections, which could lead to missed optimal routes.

Finally, Safety module scans for obstacles and behaviors to trigger a binary alert. Decision trees are effective here; they are quick, clear, and suitable for yes/no decisions based on features like "is the car ahead braking suddenly?" or "is an object closer than 10 meters?" We can adjust the branches of the tree to be cautious, focusing on safety. For the CES demo, this means quick implementation. The staff can be trained on a modest dataset of threat scenarios and debug easily, as the staff can trace why an alert was triggered. Research on anomaly detection in autonomous driving supports that decision trees help classify threats like driver impairments or unusual traffic. (sciencedirect.com , driver impairment detection using decision trees based feature selection and classification, june 2023), (openaccess.thecvf.com , a vision based system for traffic anomaly detection using deep learning and decision trees)

Expert systems might struggle with too many rules in dynamic environments, like predicting erratic pedestrians. Neural networks, while powerful for finding patterns, often act as "black boxes." They are hard to explain during safety audits and could overfit to rare events, wasting time on extensive data collection.

Overall, this combination works well for autonomous driving. Neural networks offer flexibility when data is abundant. Expert systems provide reliability for structured tasks. Decision trees ensure quick, understandable safety checks. Using mismatched methods, like neural networks for safety, could lead to untraceable errors, undermining trust, especially in a demo where reliable technology must be showcased.

b)

For inputs, mix of sensors will be used to give the neural network a complete view of the surroundings and the state of the vehicle, along with the short term path from Navigation, like (turn left in 50 meters). Key sensors include:

1- Cameras (front and side facing): These capture visuals such as lane lines, traffic signs, and pedestrians. The benefits are detailed scene information, low cost, and functionality similar to human vision, which aids in complex recognition. The drawbacks are difficulties in poor weather, like fog or glare, which can confuse the system.

Data type will mostly be continuous, as raw images or extracted features, for instance, pixel intensities over time, supporting smooth processing.

2- LiDAR or Radar: These measure the distances and speeds of objects near the car. The advantages of radar include good performance in rain or darkness, providing reliable speed data, while LiDAR generates accurate 3D maps. However, LiDAR is expensive and consumes a lot of power, and radar has lower resolution for shapes. Data type will be Continuous, such as point clouds or velocity vectors, suitable for real-time adjustments without jumps.

3- GPS combined with HD map snippets: This identifies the location of the car and includes road rules like speed limits. The advantages are that it offers context for planning and is accurate in open spaces. The disadvantages are that signals can drop in tunnels or urban (canyons), leading to location errors. Data type will be a mix of continuous coordinates plus discrete map features, for example, (stop sign ahead).

4- IMU (Inertial Measurement Unit) and wheel odometry: These track the motion of the car, including acceleration, yaw rate, and wheel speed. The benefits include high speed and internal operation without relying on external sources, which is great for stability control. However, they can drift over time due to noise, needing corrections from other sensors. Data type will be Continuous readings, for example, acceleration in m/s², allowing for precise control.

Continuous data overall for inputs is favorable because it helps the neural network find subtle patterns, such as gradual speed changes, leading to smoother driving. Discrete data, such as binary (obstacle yes/no), is simpler but could cause sudden reactions; it will be used sparingly, for example, for map labels. The sensor overviews in autonomous vehicles highlight the importance of combining these for reliability and point out how the challenges cameras face in bad weather are balanced by the reliability of radar. (foresightauto.com , april 3, 2023), (mdpi.com , An overview of autonomous vehicles sensors and their vulnerability to weather conditions)

For outputs, the Controller should produce signals that directly command the actuators of the car: steering angle (how much to turn the wheel), throttle percentage (gas pedal level for acceleration), and brake pressure (how hard to stop). These are mostly continuous values, like steering from -30° to +30° in small increments, which leads to comfortable and efficient rides. discrete outputs will be included like gear selection (forward/reverse) or turn indicators (on/off) for specific actions. Continuous outputs are preferred because driving is complex; a small adjustment to the throttle can prevent jerky starts, enhancing fuel efficiency and passenger comfort. The advantages of continuous outputs are precision and adaptability, while the disadvantage is that they require more processing power for real time calculations. In one year demonstration, this setup ensures smooth operation and drawing from actual AV control systems.

c)

The Safety module signals danger as (ON) and a clear state as (OFF). This system is important but can fail in two ways.

1- False positives (FP) can trigger alarms when there is no real threat, like mistaking a plastic bag for a pedestrian.

2- False negatives (FN) occur when a real threat is missed, such as overlooking a swerving car.

To minimise both types of errors, smart feature design and sensor fusion should be started with. This can be achieved by combining camera images with radar distances to verify threats. This method can reduce confusion that leads to False Positives, like ignoring shadows, and False Negatives, such as failing to detect hidden objects.

When creating decision trees, thresholds should be adjusted carefully. Lower thresholds make the system more sensitive, resulting in fewer False Negatives but more False Positives. Higher thresholds produce the opposite results. This would be tested during simulated drives to find the right balance. Implement class balanced training by giving more weight to rare threat examples. This helps decrease False Negatives without increasing False Positives. temporal smoothing should be used to ensure a threat only triggers an alert if it lasts over several frames. because this filters out brief False Positives caused by glitches. Finally, fail-safes should be included.because If the system is uncertain, it should default to slowing down, which helps manage any missed False Negatives. Research on AV safety highlights the importance of these strategies, including ensemble methods to improve precision and recall.

These errors affect reliability in different ways. False Positives can undermine user trust. Constant false alarms can lead drivers to ignore real dangers, causing frustration and unnecessary stops. This can lower efficiency, such as causing traffic jams from sudden braking. for CES demo, too many False Positives could make the car seem unreliable and deter potential investors. However, False Negatives are even more concerning. Missing a threat could lead to collisions, injuries, or fatalities, harming safety claims and possibly resulting in lawsuits. They can damage the overall credibility of the system since one missed threat might suggest the car is not safe enough. Reducing False Negatives would be focused on by biasing the decision tree toward caution while accepting some False Positives, which could be balanced with quick rechecks and clear explanations on the dashboard. Studies show that unbalanced errors, particularly False Negatives, create significant risks in AI driven vehicles.

Task 3 – Data collection and training pipeline

a)

Data collection for Deep Drive will be as the following:

1- Public datasets: The comma2k19 dataset (Harnett et al., 2019) delivers 33 h of highway driving (around 2 TB) captured with six cameras, one LiDAR and CAN bus steering/throttle data (https://github.com/commaai/comma2k19) , (arxiv.org, A Commute in Data: The comma2k19 Dataset). Using this source eliminates the need for an initial fleet deployment and saves a good amount of money in sensor hour

2- Targeted data gaps: Low speed urban maneuvers and construction-zone scenes are undersampled in comma2k19. A single 4×4 test truck will log an additional 120 h (approximately 8 TB) over eight weekends; vehicle rental, fuel and driver cost around $4 200.

3- recollection over time: A quarterly recollection cycle is planned to capture seasonal weather variation and road-work changes. Budgeting $1 000 per quarter keeps the annual cost ≤ $4 000.

Comma2k19 is already synchronised and timestamped; semantic segmentation masks for 200 k frames are crowd-labelled via Amazon SageMaker Ground Truth at $0.08 per object mask, totalling around $3,400 (Amazon Web Services, 2025).

Domain specific labels (lane-change intent, construction-barrel class) are added by two in-house annotators working 60 h each at $25 h^{-1} ($3 000). Active-learning uncertainty sampling (Settles, 2012) reduces subsequent labelling effort by 35 %.

Following the autonomous driving convention of Bojarski et al. (2016), the aggregated corpus is split 70 %/15 %/15 % into training/validation/test sets stratified by geographical region to avoid location bias. Temporal stratification is applied to prevent leakage across re-collection cycles.

b)

The controller network, which includes a ResNet-34 backbone and GRU, is trained using the Adam optimizer for 150 epochs. With a single NVIDIA RTX 4090 that has 24 GB of VRAM, the training time is 38 hours (Leos Developments, 2023) . If deep drive uses Google Cloud's a2-highgpu-1g with an A100 (40 GB), the cost is $3.67 per hour, totaling about $140 for a full run (Google Cloud, 2025), (gcloud-compute.com, 2025).

For low speed maneuvers, RL fine tuning with PPO takes place in the Isaac Gym simulator on an on premise Intel i9-13900K workstation. Deep Drive would complete 2 million environment steps in 22 hours without any cloud expenses.

Supervised training ends when the moving average validation loss does not improve by more than 0.5% over 10 epochs; early stopping patience is set at 10. PPO training stops when the episodic return levels off within one standard deviation for 1 million steps (Schulman et al., 2017).

Deep Drive would use the validation set to track the mean absolute error for steering angle and log loss for safety module decisions. If a gap widens between training and validation errors, it indicates overfitting. This prompts an increase in the dropout rate from 0.3 to 0.5 and the L2 regularization coefficient from 1e-4 to 5e-4. On the other hand, if both training and validation losses stagnate, it means we are underfitting. We fix this by reducing weight decay and increasing the learning rate from 1e-4 to 5e-4.

A closed loop Hardware in the Loop rig plays back 50 hours of unseen test set sequences through the actual ECU. We report performance as follows:

1. Controller: RMS lateral deviation is ≤ 0.25 m at 80 km/h (Krajewski et al., 2018).
2. Safety module: Area under ROC is ≥ 0.95 with a false negative rate of ≤ 2%.

Task 4 – Deployment

Heavy rain or drifting debris can trigger LiDAR ghosts, which lead to unnecessary braking. To prevent this, a multi sensor Bayesian fusion system ignores isolated LiDAR returns that do not have confirmation from radar or camera (mdpi.com , Exploring the Unseen: A Survey of Multi-Sensor Fusion and the Role of Explainable AI (XAI) in Autonomous Vehicles).

A temporal hysteresis filter holds back alerts unless a detection lasts for three consecutive frames, each lasting 25 ms. If uncertainty remains, the controller defaults to a 0.2 g deceleration and activates the hazard lights. This method has been tested in the nuScenes adverse weather set (Caesar et al., 2020).

GNSS dropout in tunnels can cause map localization drift, risking lane departures. To safeguard against this, a dead reckoning module combines wheel encoder and IMU data to maintain position for up to 15 seconds with less than 1 meter of error (Zhang and Singh, 2018). When the system detects the tunnel exit through the reappearance of the camera horizon, it resets the position using Monte Carlo localization against a pre loaded HD map (Hess et al., 2016). If lateral deviation exceeds 0.5 meters, steering control is limited to 2 degrees per second to ensure passenger comfort.

Temporary (detour) signs that differ from the training data might be overlooked by the vision network. To handle this, a rule based expert system monitors CAN bus turn signal requests from the safety driver. If the driver signals a turn that contradicts the planned route, the navigation layer starts a rerouting request using real time traffic APIs (OpenStreetMap, 2025). Meanwhile, the controller reduces speed to 30 km/h and increases lateral safety margins to 1.5 meters until the new route is confirmed.

References:

Amazon Web Services (AWS) (2018) Amazon SageMaker Ground Truth automated data labeling service [Slide deck]. Available at: https://pages.awscloud.com/rs/112-TZM-766/images/2018_1201-MCL_Slide-Deck.pdf .

Amazon Web Services (AWS) (2025) Amazon SageMaker Ground Truth – Product Overview. Available at: https://aws.amazon.com/sagemaker/ai/groundtruth/ .

Bojarski, M., Del Testa, D., Dworakowski, D., Firner, B., Flepp, B., Goyal, P., Jackel, L.D., Monfort, M., Muller, U., Zhang, J., Zhang, X., Zhao, J. and Zieba, K. (2016) End to End Learning for Self-Driving Cars. arXiv:1604.07316. Available at: https://arxiv.org/abs/1604.07316 .

Caesar, H., Bankiti, V., Lang, A.H., Vora, S., Liong, V.E., Xu, Q., Krishnan, A., Pan, Y., Baldan, G. and Beijbom, O. (2020) nuScenes: A multimodal dataset for autonomous driving. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2020). Available at: https://openaccess.thecvf.com/content_CVPR_2020/papers/Caesar_nuScenes_A_Multimodal_Dataset_for_Autonomous_Driving_CVPR_2020_paper.pdf .

comma.ai (2019) comma2k19 dataset [GitHub repository]. Available at: https://github.com/commaai/comma2k19 .

Google Cloud (2025) GPU pricing and a2-highgpu family (A100). Available at: https://cloud.google.com/compute/gpus-pricing .

Hess, W., Kohler, D., Rapp, H. and Andor, D. (2016) Real-time loop closure in 2D LiDAR SLAM. In: Proceedings of the IEEE International Conference on Robotics and Automation (ICRA 2016). Available at: https://research.google/pubs/real-time-loop-closure-in-2d-lidar-slam/ .

Krajewski, R., Bock, J., Kloeker, L. and Eckstein, L. (2018) The highD dataset: A drone dataset of naturalistic vehicle trajectories on German highways for validation of highly automated driving systems. arXiv:1810.05642. Available at: https://arxiv.org/abs/1810.05642 .

MDPI (Zhang, X. et al.) (2021) Sensor and sensor fusion technology in autonomous vehicles: A review. Sensors (MDPI), 21(6), 2140. Available at: https://www.mdpi.com/1424-8220/21/6/2140 .

Patent WO2020160519A1 (2020) Methods for lidar ghost or false return suppression. Available at: https://patents.google.com/patent/WO2020160519A1/en .

Schafer, H., Santana, E., Haden, A. and Biasini, R. (2018) A commute in data: The comma2k19 dataset. arXiv:1812.05752. Available at: https://arxiv.org/abs/1812.05752 .

Schulman, J., Wolski, F., Dhariwal, P., Radford, A. and Klimov, O. (2017) Proximal Policy Optimization Algorithms. arXiv:1707.06347. Available at: https://arxiv.org/abs/1707.06347 .

Settles, B. (2009) Active learning literature survey. University of Wisconsin–Madison, Computer Sciences Technical Report 1648. Available at: https://burrsettles.com/pub/settles.activelearning.pdf (Accessed: 20 August 2025).

Thrun, S., Fox, D. and Burgard, W. (2001) Monte Carlo localization for mobile robots. In: Proceedings of the IEEE International Conference on Robotics and Automation (ICRA 2001). Available at: https://robots.stanford.edu/papers/thrun.hybrid-mcl.pdf

Zhang, J., Xu, Y., Wang, J., Chen, L., Qi, X. and Zhu, J. (2022) Multi-modal sensor fusion for auto driving perception: A survey. arXiv:2202.02703. Available at: https://arxiv.org/abs/2202.02703 .

Zhou, Y., Ma, Z., Wu, T., Luo, Y., Wang, R., Chen, X., Zhang, Y., Peng, Y., Zhu, Y., Wu, X. and Yin, S. (2023) A survey on sensor failures in autonomous vehicles. Frontiers in Neurorobotics, 17, 11360603. Available at: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11360603/ .